

4 surprising health effects of noise pollution | UCLA Health

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It should come as no surprise that exposing yourself to loud noise (chronic noise exposure) can lead to hearing loss. But chronic noise can also affect your health in other significant ways. Experts are increasingly recognizing the dangers of loud noise — not only as an environmental pollutant — but also as a serious health risk. After air pollution, noise is the second-largest environmental cause of physical and mental health problems, according to the World Health Organization. The good news is that once you understand the health impact of the noises you hear every day, you can take action to protect yourself. What is chronic noise exposure? Noise pollution is any disturbing sound that affects your health and well-being. However, even if noise isn't bothering you, it could still have health impacts. Sound levels are measured in decibels (dBA), and experts agree that noise measuring 85 dBA or higher is considered harmful. To put it in perspective, dBA ranges for common sounds include: - Typical office noise: 60 dBA - Vacuum cleaner: 70 dBA - Heavy traffic: 80 to 89 dBA - Lawn mowers and motorcycles: 90 to 99 dBA - Sports event or rock concert: 120 to 129 dBA Listening to sounds at 85 dBA or more for more than eight hours at a time can lead to hearing loss, and that time decreases as noise level increases. But when it comes to other health effects, the evidence is less clear. While no one has pinpointed a direct relationship between time exposed and other health issues, experts tend to agree that long-term noise exposure can do damage. Here are signs that you are in — or just left — a setting with an unhealthy noise level: - You need to raise your voice to be heard. - You cannot hear someone who is 3 feet away. - Sounds seem muffled after leaving a noisy place. - You have pain or ringing in your ears (tinnitus) that lasts anywhere from a few days to a few weeks. Chronic noise exposure and your health Emerging evidence continues to show the impact of noise pollution on many aspects of human health. Long-term exposure to noise may: 1. Affect stress hormones When you are exposed to noise pollution, your body activates its hypothalamic-pituitary-adrenal (HPA) axis and sympathetic nervous system — two critical systems involved in the stress response. The HPA axis regulates the release of cortisol (stress hormone), and the sympathetic nervous system triggers the “fight-or-flight” feeling you get in times of stress. Ongoing stress can increase your risk for chronic inflammation and several diseases. Unlike noise-induced hearing loss (NIHL), which results from exposure to loud noise, your body's stress response can kick in with any level of chronic noise. It doesn't matter whether you have low-level traffic noise in the background or are always at loud concerts. Your brain will respond in a similar way to either level of chronic noise. 2. Increase risk of heart disease Research shows that ongoing, high-noise exposure from various sources can lead to a range of cardiovascular diseases — and the more you're exposed, the higher your risk. However, the most common cardiovascular effect of noise pollution is elevated blood pressure. Blood pressure can rise as a result of your body's stress response — especially when

loud noises trigger stress and put you in “fight-or-flight” mode. Constantly switching from low to high noise levels can also be harmful, even if the average decibel level seems healthy. So, if you live in a quiet town but down the street from a firehouse, the intermittent fire sirens may be just as damaging as living in an area where it’s always moderately noisy. The Center for Occupational and Environmental Health reports that exposure to loud, sudden sounds against a typically quiet background can be associated with heart disease, heart attacks, heart failure and stroke.

3. Lead to mental health issues When chronic exposure to noise increases stress hormones and affects your central nervous system, the resulting inflammatory state may increase your risk for mental health conditions including anxiety, depression and behavioral issues. One review of previous research reports that several studies associate road traffic noise exposure with mental health. It can increase the risk of depression by 4% and anxiety by 9%.

4. Cause cognitive impairment Over time, exposure to any noise — even typical residential noise — may affect your cognition. Recent evidence shows that living with higher levels of residential noise (averaging 50 to 75 dBA) is directly associated with cognitive impairment in people age 45 and older. Other research shows that constant low-frequency noise — coming from common things such as air conditioners, fans, traffic and construction — can be harmful. Long-term exposure to low-frequency noise may negatively impact some higher-order cognitive functions, including logical reasoning, mathematical calculation and data processing.

Protect yourself from noise pollution According to the American Speech-Language-Hearing Association, these steps may safeguard your ears and health from noise:

- Check noise ratings on appliances, power tools and hair dryers and purchase quieter products.
- Give your ears a break, by choosing not to listen to loud sounds for long.
- Lower the volume by keeping personal devices set to half volume and asking local business owners to turn down the sound in movie theaters, bars and fitness clubs.
- Wear protection, such as earplugs or earmuffs, which can cut noise by up to 30 dBA.